Week 3

1-Number of Zeros in a Given Array

```
#include <stdio.h>

int zero(int a[], int l, int h) {
    if (l > h) return 0;
    if (a[l] == 0) return h - l + 1;

    int m = (l + h) / 2;

    if (a[m] == 1)
        return zero(a, m + 1, h);
    return zero(a, l, m);
}

int main() {
    int n, a[100];
    scanf("%d", &n);

    for(int i=0; i<n; i++)
        scanf("%d", &a[i]);

    printf("%d", zero(a, 0, n-1));
    return 0;
}
```

2-Majority Element

```
#include <stdio.h>

int main() {
    int n, a[50000], count = 0, candidate;

    scanf("%d", &n);

    for(int i = 0; i < n; i++)
        scanf("%d", &a[i]);

    for(int i = 0; i < n; i++) {
        if(count == 0)
            candidate = a[i];

        if(a[i] == candidate)
            count++;
        else
            count--;
    }

    printf("%d", candidate);

    return 0;
}
```

3-Finding Floor Value

```
#include <stdio.h>

int floorValue(int a[], int l, int h, int x)
{
    if (l > h)
        return -1;

    int mid = (l + h) / 2;

    if (a[mid] == x)
        return a[mid];

    if (a[mid] < x)
        return (mid == h) ? a[mid] :
               (a[mid + 1] > x ? a[mid] : floorValue(a, mid + 1, h, x));

    return floorValue(a, l, mid - 1, x);
}
```

```
int main()
{
    int n, a[100], x;

    scanf("%d", &n);

    for(int i = 0; i < n; i++)
        scanf("%d", &a[i]);

    scanf("%d", &x);

    printf("%d", floorValue(a, 0, n - 1, x));

    return 0;
}
```

4-Two Elements sum to x

```
#include <stdio.h>

int find(int a[], int l, int r, int x) {
    if(l >= r) return 0;

    if(a[l] + a[r] == x) {
        printf("%d\n%d", a[l], a[r]);
        return 1;
    }

    if(a[l] + a[r] < x)
        return find(a, l + 1, r, x);

    return find(a, l, r - 1, x);
}

int main() {
    int n, a[100], x;

    scanf("%d", &n);
    for(int i=0; i<n; i++) scanf("%d", &a[i]);
    scanf("%d", &x);

    if(!find(a, 0, n-1, x))
        printf("No");

    return 0;
}
```

5-Implementation of Quick Sort

```
#include <stdio.h>

void q(int a[],int l,int h){
    if(l>=h) return;
    int i=l,j=h,p=a[l],t;
    while(i<j){
        while(a[i]<=p && i<h)i++;
        while(a[j]>p)j--;
        if(i<j){t=a[i];a[i]=a[j];a[j]=t;}
    }
    a[l]=a[j]; a[j]=p;
    q(a,l,j-1); q(a,j+1,h);
}

int main(){
    int n,a[100];
    scanf("%d",&n);
    for(int i=0;i<n;i++)scanf("%d",&a[i]);
    q(a,0,n-1);
    for(int i=0;i<n;i++)printf("%d ",a[i]);
}
```